ARTEA: Theory-Guided Hierarchical Graph Index for High-Performance ANN Search [Technical Report]

Weitang Ye
weitang.ye@ntu.edu.sg
Nanyang Technological University
Singapore

Dingheng Mo
dingheng001@e.ntu.edu.sg
Nanyang Technological University
Singapore

Siqiang Luo
siqiang.luo@ntu.edu.sg
Nanyang Technological University
Singapore

ABSTRACT

Approximate Nearest Neighbor (ANN) search is a critical infrastructure for modern data management and AI applications. While hierarchical proximity graphs have achieved widespread empirical success due to their multi-layer, coarse-to-fine routing efficiency, they remain less understood from a theoretical perspective: can hierarchical structures be endowed with both strict accuracy guarantees and low ANN search complexity simultaneously? In this paper, we answer affirmatively by introducing Artea, a novel hierarchical proximity graph. By integrating deterministic r-net sampling with a novel Aspect-Ratio-Constrained Pruning (ARC-Pruning) technique, Artea bounds the worst-case search complexity to $O((\alpha \cdot \tau)^{\lambda} + \alpha^{\lambda} \log \Delta)$ under the practical τ-bounded scenario, where τ is a practical constant, λ is the doubling dimension of the metric space, and Δ is the dataset aspect ratio. This makes Artea the first proximity graph to achieve a strictly logarithmic dependence on Δ. Besides theoretical contributions, we further develop a robust system implementation inspired by the theoretical principles of Artea. Extensive empirical evaluations demonstrate that our approach outperforms prevailing baselines in search efficiency and result accuracy, while maintaining highly competitive index construction costs.

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The source code, data, and/or other artifacts have been made available at https://github.com/NTU-Siqiang-Group/Artea.

1 INTRODUCTION

Approximate Nearest Neighbor (ANN) search in high-dimensional spaces is a crucial infrastructure for modern data management. It has been applied in ubiquitous applications including information retrieval [17], recommendation systems [34], large language models [3, 14, 23, 25], and AI agents [36, 37]. Given a set of n data points $P \subset \mathbb{R}^d$ and a query vector $q \in \mathbb{R}^d$, our goal is to construct an index structure that allows a search algorithm to efficiently retrieve the point $p^* \in P$ closest to q. Among a diverse array of techniques to address the ANN problem, recent studies indicate that proximity graph based methods have emerged as the dominant solution, consistently demonstrating superior performance over classic approaches such as hashing techniques [12, 18, 26, 29, 43, 44], inverted indices [13, 33, 42], and tree-based structures [4, 5, 21, 22]. A *proximity graph* (PG) is a directed graph in which each vertex represents a data point, and edges are established according to *predefined criteria*, with the purpose of approximately locating the nearest neighbor of a query point by navigating the PG.

Within the family of PG-based methods [2, 9–11, 16, 24, 28, 30, 31, 35, 38, 41, 45, 46], several pioneering works [11, 19, 24, 28, 38] have conducted in-depth investigations into theoretically sound proximity graphs, as summarized in Table 1. A predominant paradigm among these solutions is employing pruning techniques to construct sparse graphs that support accurate ANN search through greedy routing. Intuitively, edge-pruning techniques start from a complete graph and prune a directed edge (u, v) if there exists another neighbor w of u that is sufficiently close to v, rendering the direct connection (u, v) redundant. This procedure is deterministic, as candidates are consistently processed in ascending order of their distance to u. Depending on the specific criteria used to define whether w and v are sufficiently close, various pruning strategies have emerged [11, 19, 24, 38], offering distinct trade-offs between search accuracy and query complexity. Among these methods, α-CG [24] represents the state of the art, achieving an exact retrieval guarantee with *polylogarithmic time complexity* under the practical setting where $\delta(p^*, q)$ is bounded. However, an inherent limitation of this approach is that its pruning process requires evaluating candidate vertices across the entire dataset. This global scope drastically inflates the per-step routing cost to $O(\log \Delta)$, where the dataset aspect ratio Δ acts as a scale-dependent variable [1], which describes the ratio between the maximum and minimum pairwise distance. Most recently, Lu and Tao [28] proposed an approach that exploits the spatial properties of a specific data structure (i.e., r-nets [22]) to accelerate graph construction. Nevertheless, this design still incurs an $O(\log \Delta)$ cost per routing step, leaving the overall query complexity with a polylogarithmic dependence on Δ.

In practice, these theoretically rigorous prototypes often incur prohibitively high construction costs [11, 24, 38, 41] or remain purely conceptual without implementations [28]. Consequently, these works typically introduce practical variants—heuristic approximations designed to translate these theoretical models into engineering-viable solutions. For instance, MRNG is approximated by the Navigating Spreading-out Graph (NSG), and the rigorous α-CG is relaxed to form the α-Convergent Neighborhood Graph

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[1] In practice, Δ tends to be large: on `glove-100d` dataset, for example, we measured $\Delta \approx 193316$ using Faiss L2, which makes the $O(\log \Delta)$ factor a meaningful cost.

(α-CNG). We also summarize the correspondence between these theoretical prototypes and their practical variants in Table 1.

Motivation. Theoretically sound, state-of-the-art indices [19, 24, 28, 38] have demonstrated strong efficiency in minimizing the number of routing steps required to locate the nearest neighbors. Their rapid search convergence is achieved by enforcing a proportional reduction in distance toward the query target at every hop. However, to guarantee such a substantial distance reduction property, the construction algorithm must evaluate candidate neighbors spanning the entire dataset when applying the tailored pruning inequalities. Consequently, the maximum out-degree of a vertex becomes inevitably entangled with $\log \Delta$, where Δ represents the dataset's aspect ratio (the ratio of the dataset's diameter to its minimum pairwise distance). This leads to inefficiency in index construction and the total search time to be at least $O((\log \Delta)^2)$.

Therefore, our core intuition is to structurally decompose a single expensive $O(\log \Delta)$ routing step into a sequence of lightweight, $O(1)$-cost "mini-steps". Executing a constant number of these mini-steps guarantees the same strict exponential decay in distance to the query, thereby bounding the total search time to $O(\log \Delta)$.

Solution. To realize the aforementioned theoretical properties, we adopt a hierarchical graph architecture. Traditional multi-tiered structures (such as HNSW) typically rely on random sampling to assign points to different layers. Instead, we aim to systematically co-design both the vertex hierarchy and the edge topology across the hierarchy to guarantee optimal search behaviors. Specifically, we deterministically populate the layers by constructing r-nets in a bottom-up fashion, securing a mathematically rigorous vertex distribution at each level. On top of this geometrically robust vertex hierarchy, we build the intra-layer proximity graphs using our novel ARC-Pruning (Aspect-Ratio-Constrained Pruning) technique. By restricting the aspect ratio of every vertex's candidate neighbor list, ARC-Pruning effectively limits per-layer out-degree of every vertex to a constant, ensuring that the cost of intra-layer routing remains $O(1)$ while preserving the critical navigability required to achieve exponential distance decay.

By integrating the deterministic bottom-up r-net hierarchy with ARC-Pruning, our proposed structure, Artea, successfully advances the state-of-the-art in proximity graphs by reducing the search time complexity's dependence on the dataset aspect ratio Δ from polylogarithmic to strictly logarithmic. As summarized in Table 1, we present a rigorous theoretical comparison between Artea and prevailing PG-based methods under the practical τ-bounded scenario (τ is a practical constant, following [24, 38]). Crucially, Artea is the first proximity graph to achieve *a strictly logarithmic dependence on* Δ in search complexity under τ-bounded scenario and doubling metrics, surpassing all state-of-the-art flat indices. To further refine this bound, decoupling the pruning strategies between the upper and bottom layers allows Artea to transform the τ^λ term (λ is the doubling dimension which is treated as a constant) from a coupled multiplier into an isolated additive factor, yielding a worst-case search complexity of $O\left((\alpha \cdot \tau)^\lambda + \alpha^\lambda \log \Delta\right)$.

Finally, in line with existing works such as NSG [11], Vamana [41], τ-MNG [38] and α-CNG [24], we introduce efficient approximations that enable the rapid construction of the Artea graph while retaining its structural advantages. Extensive empirical evaluations demonstrate that this practical implementation achieves significant improvements in both indexing speed and query throughput, decisively outperforming existing state-of-the-art solutions.

Contributions. The contributions of this paper are as follows:

- **Artea: A Principled Hierarchical Architecture.** We introduce Artea, which systematically co-designs vertex distribution and edge topology. By integrating a deterministic r-net sampling with our novel ARC-Pruning, Artea strictly limits the per-layer out-degree to a constant. This ensures $O(1)$ intra-layer routing costs while preserving optimal global navigability.
- **Rigorous Theoretical Guarantees.** Artea stands as the first proximity graph to achieve a strict logarithmic search dependence on Δ, under the practical τ-bounded scenario. By isolating the τ^λ overhead into an additive factor, we bound the worst-case complexity to $O\left((\alpha \cdot \tau)^\lambda + \alpha^\lambda \log \Delta\right)$, asymptotically better than state-of-the-art proximity graphs.
- **Robust System Implementation.** Bridging theory and practice, we implement Artea from scratch in over 9,000 lines of `C++` template code. Our implementation centers on a decoupled construction framework—incremental insertion for the sparse upper layers and a refinement engine for the massive bottom layer—whose high-performance infrastructure supports fast index construction.
- **Superior Empirical Performance.** We conduct extensive experiments on multiple real-world datasets to evaluate the practical search performance of Artea. Our empirical results demonstrate that Artea substantially outperforms prevailing flat and hierarchical baselines.

2 PRELIMINARIES

This section defines the ANN search problem and introduces the notations used throughout our analysis. Then, we provide an overview of Proximity Graphs and formulate the generalized Hierarchical Greedy Search algorithm.

2.1 Problem Definition and Notation

A metric space $(\mathcal{M}, \delta)$ is a set of points $\mathcal{M}$ endowed with a distance function $\delta: \mathcal{M} \times \mathcal{M} \to \mathbb{R}_{\geq 0}$, which satisfies the triangle inequality $\delta(u, w) \leq \delta(u, v) + \delta(v, w)$ for any $u, v, w \in \mathcal{M}$. The distance between a point p and a set S is calculated by $\delta(p, S) = \min_{v \in S} \delta(p, v)$.

Consider a database $P \subseteq \mathcal{M}$ consisting of n points in d-dimensional space. For a query point q in $\mathcal{M}$, a point $p^* \in P$ is a ***nearest neighbor*** (NN) of q if $\delta(p^*, q) = \delta(q, P)$; while a point $p \in P$ is a $(1+\epsilon)$-***approximate nearest neighbor*** ($(1+\epsilon)$-ANN) of q for some constant $\epsilon > 0$ if $\delta(p, q) \leq (1+\epsilon) \cdot \delta(p^*, q)$. Our objective is to develop a data structure that efficiently solves the ANN search problem with the aforementioned accuracy guarantee.

This paper focuses on the threshold-constrained setting where $\delta(q, P) \leq \tau \cdot \rho$ for a user-defined constant τ, where ρ denotes the minimum pairwise distance in dataset P. This is motivated by the observation that in real-world benchmark workloads, queries are typically distinct from data points in P yet remain spatially close to their nearest neighbors [24, 38].

We note that in real-world scenarios, empirical studies often employ ranking-based metrics, such as *recall*, to compare the returned

Table 1: Comparison of PG-based methods (under practical $\delta(q, v^*) < \tau\rho$ setting). The notations follow from [24].

Theoretical Prototype	Accuracy Guarantee	Routing Property	Average Complexity	Worst-case Complexity	Approximate Variant	Practical Performance‡	
						Querying	Construction
MRNG [11]	☑	Monotonic Path	$O(n^{\frac{2}{d}} \ln n)$	$O(n)$	NSG	1.00×	1.00×
Vamana [19]	✓	α-Shortcut Reachable	–	$O\left(\alpha^\lambda \log \Delta \log \frac{\Delta}{(\alpha-1)\epsilon}\right) \overset{\dagger}{\approx} O(\log^2 \Delta)$	Vamana [41]	1.60×	4.35×
G_{net} [28]	✓	Log Drop	–	$O\left((1/\epsilon)^\lambda \cdot \log^2 \Delta\right) \overset{\dagger}{\approx} O(\log^2 \Delta)$	–	–	–
τ-MG [38]	✔	τ-Monotonic Path	$O(n^{\frac{1}{d}} (\ln n)^2)$	$O(n)$	τ-MNG	1.39×	1.05×
α-CG [24]	✔	α-Reducible	–	$O\left((\alpha \cdot \tau)^\lambda \log \Delta \log_\alpha \Delta\right) \overset{\dagger}{\approx} O(\log^2 \Delta)$	α-CNG	1.68×	0.55×
ARTEA (Ours)	✔	Hierarchical Drop	–	$O\left((\alpha \cdot \tau)^\lambda + \alpha^\lambda \log \Delta\right) \overset{\dagger}{\approx} O(\log \Delta)$	**ARTEALIB**	3.31×	4.62×

✔: exact; ✓: $\left(\frac{\alpha+1}{\alpha-1} + \epsilon\right)$-ANN; ☑: guaranteed only when $\delta(q, v^*) = 0$; ×: no guarantee; †: asymptotic equality up to the constants α, τ, λ; ‡: **Querying** and **construction** performance (higher is better) is averaged across all evaluated datasets and normalized to NSG (see Sec. 6 for details).

set with the true top-k results. Accordingly, $(1+\epsilon)$-ANN framework can be naturally generalized to approximate k-nearest neighbor (k-ANN) search problem by maintaining a sorted candidate list, known as *Beam Search* algorithm [11, 24, 38]. Furthermore, to analyze the complexity of Proximity Graphs (PGs), we utilize the framework of doubling metrics to quantify the intrinsic dimensionality of the data [5, 22], based on the following key notations:

Definition 2.1 (Aspect Ratio). Let $d_{\max}$ and ρ denote the diameter and the minimum pairwise distance of P, respectively. The *aspect ratio* of dataset P is defined as $\Delta = d_{\max}/\rho$.

Definition 2.2 (Doubling Dimension [5]). For any $p \in \mathcal{M}$ and $r > 0$, let $B(p, r) = \{x \in \mathcal{M} \mid \delta(p, x) < r\}$ be the radius-r ball. The *doubling dimension* of $(\mathcal{M}, \delta)$ is the smallest constant λ such that any ball $B(p, r)$ can be covered by at most 2^λ balls of radius $r/2$.

While the doubling dimension of a set $P \subseteq \mathbb{R}^d$ (equipped with a metric ℓ_p norm) is at most $\Theta(d)$, empirical studies reveal that real-world high-dimensional datasets typically possess an intrinsic dimension much lower than their ambient dimension d [19]. Following recent theoretical frameworks [19, 24, 28], we assume the doubling dimension λ of our input dataset P is a small constant, i.e., $\lambda = O(1)$. The following is a fundamental property [28]:

LEMMA 2.3. *Let $X \subset \mathbb{R}^d$ be a set with aspect ratio Δ_X. The cardinality of X is bounded by its aspect ratio, i.e., $|X| \leq (\Delta_X)^{O(\lambda)}$.*

Since the intrinsic dimension λ is bounded, this inequality implies that if any subset X has a constant aspect ratio, then it contains only a constant number of points. In contrast, the global aspect ratio Δ of the entire input dataset P is typically non-constant and treated as a growing term with dataset size in complexity analysis.

2.2 Proximity Graphs

Proximity Graphs based indices have emerged as the state-of-the-art solution for ANN search in high-dimensional vector spaces. Formally, a PG is a directed graph $\mathcal{G}(\mathcal{V}, \mathcal{E})$ where the set of vertices $\mathcal{V}$ corresponds to the original points in the database P and the edges $\mathcal{E}$ are carefully selected to facilitate the search algorithm to navigate the graph towards the nearest neighbor of q.

In the theoretical modeling of proximity graphs, an extensively studied routing procedure is ***Greedy Search*** [11, 19, 24, 28, 38], which iteratively hops from the current vertex p° to its closest out-neighbor with respect to q, terminating at a local minimum. Modern hierarchical indices [27, 31, 46] extend this paradigm into

Algorithm 1: Hierarchical Greedy Search

Input: graph $\mathcal{G}$, query point q, entry point $p^\top$, hierarchy height $h_{\max}$
Output: approximate nearest neighbor of q

```
current hop vertex p° ← p^⊤
for h from h_max to 0 do
    while true do
        N_h(p°) ← out-neighbors of p° in layer G_h
        u* ← argmin {δ(x, q) | x ∈ N_h(p°)}
        if δ(u*, q) ≥ δ(p°, q) then break;
        p° ← u*
return p°
```

Hierarchical Greedy Search (Algorithm 1). The hierarchy is organized into layers $\mathcal{G}_h$ induced over subsets $\mathcal{V}_h \subseteq \mathcal{V}_{h-1}$, with $\mathcal{G}_0$ containing the entire dataset. Starting from a global entry $p^\top$, the algorithm performs greedy routing at each level, passing the local minimum p° down as the starting point for the next. This top-down traversal guarantees a monotonically decreasing distance $\delta(p^\circ, q)$. Upper levels exploit long-range edges for rapid convergence, while $\mathcal{G}_0$ performs fine-grained refinement.

Intuitively, The query efficiency of this process is dominated by the number of distance computations of each step (governed by vertex out-degrees) and the number of hops required to converge [11, 24, 38]. These two factors exhibit an intrinsic trade-off: augmenting the graph's out-degree generally decreases the number of routing steps (e.g., in the extreme case of a complete graph, any destination is reachable in a single hop), but this comes at the expense of increased per-step computational overhead. Thus, the search complexity behaves as a convex function of the graph degree, and deriving an optimal configuration becomes non-trivial.

3 ARTEA GRAPH

In this section, we present the Artea Graph, a novel proximity graph that provides provable query efficiency and accuracy guarantees. We first describe its hierarchical structure in Section 3.1, then introduce ARC-Pruning in Section 3.2, and finally analyze its space usage and construction cost in Section 3.3.

3.1 Hierarchical Graph Structure

The *Artea Graph* is organized as a hierarchy of proximity graphs $\mathcal{G}$, defined as:

$$\mathcal{G} = \langle \mathcal{G}_0, \mathcal{G}_1, \ldots, \mathcal{G}_{h_{\max}-1}, \mathcal{G}_{h_{\max}} \rangle$$

where each layer $\mathcal{G}_h = (\mathcal{V}_h, \mathcal{E}_h)$ represents a distinct graph with vertex set $\mathcal{V}_h$ and edge set $\mathcal{E}_h$. Consistent with state-of-the-art hierarchical indexing schemes [31, 46], vertices for level h are selected from the preceding layer $\mathcal{V}_{h-1}$. This raises the following design question: *how should each layer $\mathcal{V}_h \subset \mathcal{V}_{h-1}$ be selected so that the resulting hierarchy supports the most effective navigation possible?*

Ideally, to function as an efficient navigational backbone, the upper-level vertices should be uniformly distributed to cover the underlying space. Popular hierarchical graph indices, such as HNSW [31] and MIRAGE [46], typically rely on randomized vertex promotion, where vertices are sampled into upper layers with a fixed probability. However, such oblivious probabilistic selection lacks deterministic spatial guarantees, exposing the hierarchy to two fundamental structural inefficiencies:

Navigational blind spots: A local spatial region may fail to promote any representative vertex to the upper layer. The absence of an upper-level anchor forces the coarse-to-fine routing process to rely on suboptimal, circuitous paths to approach targets in that unrepresented region, which severely degrades query efficiency.

Redundant clustering: Conversely, the randomized strategy may inadvertently promote multiple vertices within tight proximity. This spatial redundancy compromises the sparsity of the upper layers, preventing the graph from maintaining a low-degree topology essential for rapid, long-range metric space traversal.

To explicitly circumvent these structural pitfalls, Artea Graph employs a deterministic selection strategy based on r-net [22], a fundamental tool in metric spaces, defined as follows:

Definition 3.1 (r-Net Selection Strategy). Given a subset $X \subseteq \mathcal{M}$ and a value $r > 0$, the *r-Net Selection Strategy* selects a subset $X_r \subseteq X$ satisfying the following two conditions:

(1) Separation property. The pairwise distance between any two distinct points in X_r is at least r;
(2) Covering property. For any point $x \in X$, there exists at least one point $x' \in X_r$ such that $\delta(x, x') \le r$.

Let $\beta > 1$ be a constant, and define the scale sequence $\{R_h\}_{h=0}^{h_{\max}}$ by $R_h = \beta^h \rho$, where $h_{\max} = \lceil \log_\beta \Delta \rceil + 1$. Artea constructs its hierarchy recursively as follows: the base layer is $\mathcal{V}_0 = P$, and for each level $h \ge 1$, the vertex set $\mathcal{V}_h$ is chosen as an R_h-net of $\mathcal{V}_{h-1}$. Since $R_{h_{\max}}$ exceeds the diameter of P, the top layer contains only one point. A direct consequence of this recursive construction is the *hierarchical covering property*, which guarantees that every point retains a nearby representative at every level of the hierarchy.

Lemma 3.2 (Hierarchical Covering). *For any point $p \in P$ and any level $h \in \{0, 1, \ldots, h_{\max}\}$, there exists a vertex $u_h \in \mathcal{V}_h$ such that*

$$\delta(u_h, p) \;\le\; \sum_{k=0}^{h} R_k \;=\; \rho \sum_{k=0}^{h} \beta^k. \tag{1}$$

The bound follows by recursively applying the covering condition of r-nets along the hierarchy [22]. As we show in Section 4.2, this property is the key structural ingredient that enables logarithmic-time navigation, which justifies our choice of the r-Net Selection Strategy over the randomized sampling used by HNSW and MIRAGE. Meanwhile, because exact r-net construction can be expensive, in Section 5 we present an efficient approximation algorithm.

Algorithm 2: ARC-Pruning

Input: point p, candidate set C_h, minimum pairwise distance R_h, aspect ratio threshold $\Delta_h(\alpha, \beta, \tau, \theta)$, conflicting ratio function $\mathcal{F}_{\text{AG}}{:}\mathcal{M}\times\mathcal{M}\to\mathbb{R}$
Output: a retained neighbor set $\mathcal{N}_h(p)$ of p at level h

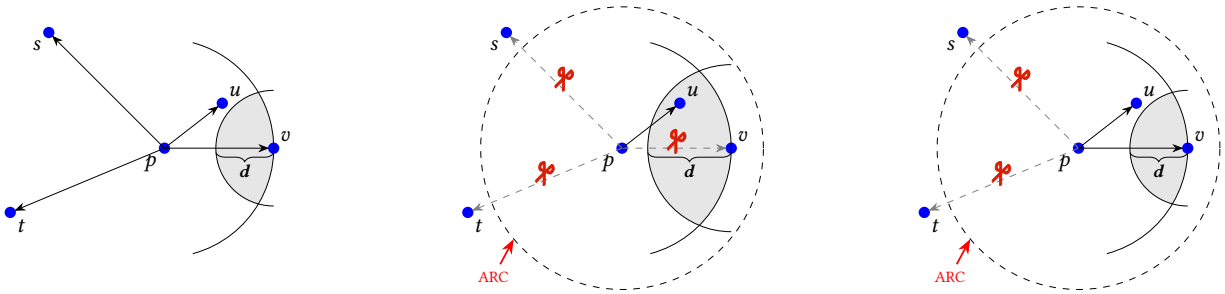
```
ARC-candidate set
  C̃_h ← {v ∈ C_h | δ(p,v) ≤ R_h · Δ_h(α,β,τ,θ)}
sort C̃_h in ascending order of the distance to p
retained neighbor set N_h(p) ← ∅
for each u ∈ C̃_h (sorted) do
    flag ← true
    for each v ∈ N_h(p) do
        if v ∈ B(u, F_AG(p,u,h)) then
            flag ← false
            break
    if flag = true then
        N_h(p) ← N_h(p) ∪ {u}
return N_h(p)
```

Having established the vertex sets for each level, we next construct the edges. Consistent with recent proximity graphs that offer theoretical guarantees, Artea Graph employs a specialized pruning strategy to determine out-edges for each vertex.

3.2 Aspect Ratio-Constrained Pruning

Recall that in the construction of proximity graphs, a critical step is to determine the connectivity for each vertex. In this task, the key is to retain shortcut vertices that navigate the search significantly closer to the query node. Flat indices such as α-CG [24] achieve this with theoretical guarantees through conservative pruning rules that preserve the edges needed for strict distance reduction. As illustrated in Figure 1a, however, this also leads to the retention of extra edges, including long-range outliers such as s and t, as well as points such as v that the inherent form of α-CG's pruning condition fails to exclude. Revisiting the pruning condition under Artea's hierarchical setting, however, yields a striking observation: the per-vertex edge budget can be drastically reduced — driving the maximum out-degree at every layer down to a constant. Two complementary mechanisms, both motivated by the hierarchical structure, together achieve this.

Mechanism 1 (Aspect Ratio Constraint): By examining the hierarchical routing process, we observe that the graph's *Hierarchical Covering Property* (Lemma 3.2) naturally bounds the local search space. Suppose the routing at the upper layer (h+1) successfully returns a valid approximate nearest neighbor, which then serves as the entry point for the current layer h. The covering property guarantees that the distance between this entry point and the target point in layer h is strictly bounded by a constant multiple of the current layer scale R_h. Consequently, this indicates that there exists a constant multiplier Δ_h, such that the entire single-layer routing path at level h is strictly confined within a local radius of $\Delta_h R_h$. Therefore, candidates outside this bounded range are strictly unnecessary for routing. By safely discarding these outliers, we bound the local candidate pool independently of the dataset's global aspect ratio Δ.



$d = \frac{1}{\alpha}(\delta(p,v) - (\alpha+1)\tau\rho)$

(a) A typical case in the pruning methods of SOTA PG (i.e., α-CG [24]).

$d = \frac{1}{\alpha}\delta(p,v)$

(b) Aggressive ARC-Pruning for upper levels ($h \geq 1$) in Artea.

$d = \frac{1}{\alpha}(\delta(p,v) - (\alpha+1)\tau\rho)$

(c) Conservative ARC-Pruning for the bottom level ($h = 0$) in Artea.

Figure 1: An Illustration of Pruning Strategies. (a) SOTA flat indices (i.e., α-CG) use a conservative pruning condition to ensure strict distance reduction and exact query guarantee, retaining both vertex v and more distant vertices s, t. (b) In Artea Graph, at upper levels ($h \geq 1$), ARC-Pruning enforces a larger conflicting radius (gray area) to aggressively prune v for graph sparsity, while the aspect ratio constraint (dashed boundary) prunes s and t. (c) At the bottom level ($h = 0$), the conflicting radius is reduced to preserve critical edges like v for exact guarantee, while still filtering out s and t.

Rather than evaluating the entire vertex set $\mathcal{V}_h$ as candidates—which inherently carries the large global aspect ratio Δ of the dataset—ARC-Pruning directly starts with a restricted candidate set. Specifically, at Line 1 of Algorithm 2, it filters the vertices right from the start by enforcing the local aspect ratio threshold Δ_h, mathematically defined as:

$$\Delta_h(\alpha,\beta,\tau,\theta) = \left(\frac{2\alpha}{\alpha-1}+\theta\right)\psi_h + \left(\frac{\alpha+1}{\alpha-1}+\theta\right)\beta. \tag{2}$$

$$\text{where } \psi_h(\beta,\tau) = \frac{\tau}{\beta^h} + \sum_{k=0}^{h}\beta^{-k},$$

Here, β is the expansion base of the hierarchy, τ is the threshold parameter in the τ-bounded setting, while θ and α are system constants related to hierarchical routing and pruning, respectively.

Using this threshold, ARC-Pruning effectively confines the candidates to a local region. As shown by the dashed boundary in Figures 1b and 1c, any candidates falling outside this restricted range, such as s and t, are universally pruned. Since $\beta > 1$ is the fixed hierarchy expansion base, the geometric series $\sum_{k=0}^{h}\beta^{-k}$ is strictly bounded by its infinite sum $\beta/(\beta-1)$. This gives $\psi_h < \tau+\beta/(\beta-1) = O(\tau+1)$. Substituting this into Δ_h, we have

$$\Delta_h < \left(\frac{2\alpha}{\alpha-1}+\theta\right)\cdot\left(\tau+\frac{\beta}{\beta-1}\right) + \left(\frac{\alpha+1}{\alpha-1}+\theta\right)\cdot\beta = O(\beta+\tau).$$

Thus, Δ_h depends only on the system parameters β, τ, and θ, rather than on the global aspect ratio Δ or the dataset size n. In the following, we write $\psi_h(\beta,\tau)$ and $\Delta_h(\alpha,\beta,\tau,\theta)$ simply as ψ_h and Δ_h whenever the context is clear.

Mechanism 2 (Layer-wise Decoupled Pruning): Furthermore, considering that we only need to perform a high-precision search at the bottommost layer to locate the exact nearest neighbor, we can explicitly differentiate the pruning strategies between the upper and lower layers. For upper layers ($h \geq 1$), the pruning can be more aggressive, allowing the search to rapidly navigate into the local vicinity of a query point. Conversely, for the bottom layer ($h = 0$), the pruning must be more conservative, preserving enough edges to guarantee exact retrieval.

Lines 2 to 11 in Algorithm 2 specify this procedure. The algorithm first sorts the ARC-candidate set $\widetilde{C}_h$ in ascending order of distance to p. It then scans each candidate u and checks whether any previously retained neighbor lies within the conflicting radius of u (Line 7). This conflicting radius is defined by Eq. (3), with scaling parameter $\alpha = (c+\theta)/\theta$, where c is a constant:

$$\mathcal{F}_{\text{AG}}(p,u,h) = \begin{cases} \delta(p,u)/\alpha & \text{if } h \geq 1; \\ \frac{1}{\alpha}(\delta(p,u) - (\alpha+1)\tau\rho) & \text{if } h = 0. \end{cases} \tag{3}$$

This level-dependent radius elegantly trade-offs sparsity and precision. For upper levels ($h \geq 1$), ARC-Pruning uses the larger threshold $\delta(p,u)/\alpha$. This prunes candidates more aggressively and produces a sparser graph, as shown in Figure 1b. Such sparsity is sufficient because upper layers only need to find high-quality entry points. At the bottom level ($h = 0$), the bias term $(\alpha+1)\tau\rho$ reduces the conflicting radius, as shown in Figure 1c. This preserves more short-range edges, such as v, which are crucial for fine-grained exploration and exact nearest-neighbor convergence.

By decoupling the pruning logic across layers, we restrict the impact of the τ multiplier—which inflates the degree complexity in flat indices like α-CG—strictly to the bottommost layer. As will be demonstrated in the subsequent complexity analysis, this strategic confinement ultimately transforms τ from a global multiplier into an additive term in the query time. The following proposition clearly illustrates how this decoupling manifests in the out-degree bounds between the upper and bottom layers:

Proposition 3.3. *The out-degree of any vertex p in layer h, denoted by $|\mathcal{N}_h(p)|$, is bounded as follows:*

$$|\mathcal{N}_h(p)| \leq \begin{cases} O\left(\alpha^\lambda \log \Delta_h\right) & \text{if } h \geq 1 \\ O\left((\alpha\tau)^\lambda + \alpha^\lambda \log \Delta_0\right) & \text{if } h = 0 \end{cases}$$

Proof Sketch. We adopt the same geometric argument as the degree analyses of DiskANN and α-CG [19, 24], centering a sequence of concentric annuli at p. Since ARC-Pruning keeps neighbors within each annulus separated in proportion to their distance from p, Lemma 2.3 bounds each annulus to only $O(\alpha^\lambda)$ neighbors, and summing over the annuli yields a bound logarithmic in the

local search range Δ_h. The crucial difference from [19, 24] is that, instead of letting this range span the entire dataset, ARC-Pruning caps the candidate aspect ratio at $\Delta_h = O(\beta + \tau)$; the out-degree is therefore governed by the local constant Δ_h rather than the global aspect ratio Δ, which inherently **decouples** the out-degree from Δ. We provide the detailed derivation in our technical report [49]. □

3.3 Space and Construction Time

Building upon the out-degree bounds established in Section 3.2, we now analyze the overall space requirements and construction overhead of the Artea Graph.

Index Size Analysis. Based on Proposition 3.3, we analyze the total storage complexity. The index size is the cumulative cardinality of neighbor sets across all layers. With hierarchy height $h_{\max} = O(\log \Delta)$ and $|V| = n$, the total size is:

$$\mathcal{S}_{\text{AG}} \;=\; n \cdot \left(O\left((\alpha\tau)^\lambda + \alpha^\lambda \log \Delta_0\right) + \sum_{h=1}^{h_{\max}} O\left(\alpha^\lambda \log \Delta_h\right)\right)$$

Given that $\log \Delta_h$ is bounded by a constant and $h_{\max}$=$O(\log \Delta)$, the final space complexity is $O\left(n \cdot \alpha^\lambda \cdot \left(\tau^\lambda + \log \Delta\right)\right)$. This confirms the index size scales linearly with n and logarithmically with the dataset aspect ratio Δ.

Construction Time. The construction procedure consists of two phases: *hierarchical vertex selection* and *hierarchical neighbor generation*. By employing the hierarchical net construction technique from [15], the vertex selection phase requires $O(n \log n)$ time. Next, we analyze the neighbor set generation (Algorithm 2) for each vertex p. Constructing the ARC-candidate set (Line 1) takes $O(n \log \Delta)$ time across all layers. The sorting procedure in Line 2 requires $O\left(\left|\widetilde{C}_h\right| \cdot \log |\widetilde{C}_h|\right)$. By Lemma 2.3, $|\widetilde{C}_h|$ is bounded by $O((\Delta_h)^\lambda) \approx O\left((\beta + \tau)^\lambda\right)$. Finally, the pruning procedure takes $O(|\widetilde{C}_h| \cdot |\mathcal{N}_h|)$. Thus, the processing time for each vertex is:

$$\begin{aligned} T_c(p) &= O(n \log \Delta) + \sum_{h=0}^{h_{\max}} \left(O(|\widetilde{C}_h| \log |\widetilde{C}_h|) + O(|\widetilde{C}_h| \cdot |\mathcal{N}_h|)\right) \\ &= O(n \log \Delta) + \sum_{h=0}^{h_{\max}} O(1) = O(n \log \Delta). \end{aligned}$$

The second equality holds because the terms $|\widetilde{C}_h|$ and $|\mathcal{N}_h|$ are bounded by structural constants (α, β, τ) independent of dataset scales n and Δ. Summing over all n vertices, the total construction time complexity is dominated by the linear scan for each vertex, resulting in $T_c(\text{total}) = \sum_p T_c(p) = O(n^2 \log \Delta)$. This is comparable to that of existing methods such as Vamana, τ-MG, and α-CG, which generally require $\Omega(n^2)$ construction time. Since an $\Omega(n^2)$ construction cost can be prohibitive for large-scale datasets, we propose a more efficient, practical implementation in Section 5 that substantially accelerates the index building process while preserving the query performance of the Artea Graph.

4 THEORETICAL ANALYSIS OF QUERYING

Having presented the construction of Artea Graph, we now analyze its query behavior. The goal of this section is to show that the proposed construction supports both accurate routing and efficient search. In particular, we will show that the hierarchy and ARC-Pruning rule together guarantee exact nearest-neighbor retrieval under the τ-bounded setting, with only logarithmic dependence on the dataset aspect ratio Δ. Our analysis focuses on the Hierarchical Greedy Search framework introduced in Section 2, but with a *refined early-stopping rule*. The search proceeds from the top layer to the bottom layer. Upper layers only need to find high-quality entry points for the next layer, so they do not require full local convergence. Instead, once the routing at level h reaches a vertex v satisfying $\delta(v, q) \le \left(\frac{\alpha+1}{\alpha-1} + \theta\right) \psi_h R_h$, it descends to $\mathcal{G}_{h-1}$. This early termination rule avoids unnecessary upper-layer refinement while preserving a sufficiently good starting point for the next layer. The bottom layer then performs the final exact nearest-neighbor search.

We first prove the accuracy guarantee of this routing process in Section 4.1, and then derive its query-time complexity in Section 4.2 by combining the routing-step bound with the out-degree bound from Proposition 3.3.

4.1 Accuracy Guarantee

In this subsection, we show why Artea Graph's routing process is theoretically effective and reliable. The key observation is that whenever the current vertex is not good enough for its layer, then ARC-Pruning ensures that there is still an outgoing neighbor closer to the query. Hence, we can prove that Artea's search would never stop at a false local minima, and it eventually reaches high-quality entry points in upper layers and the exact nearest neighbor at the bottom layer.

To facilitate the proof, we first establish several lemmas that characterize the properties of our hierarchical structure under practical settings. Let q be a query point whose exact nearest neighbor $p^* \in P$ satisfies $\delta(q, p^*) \le \tau\rho$. Furthermore, let p_h^* denote the exact nearest neighbor of q within the vertex set $\mathcal{V}_h$, and let $\hat{p}_h \in \mathcal{V}_h$ denote a vertex retrieved at level h that satisfies $\delta(\hat{p}_h, q) \le \left(\frac{\alpha+1}{\alpha-1} + \theta\right) \psi_h R_h$. This vertex $\hat{p}_h$ serves as a high-quality entry point for routing in the subsequent layer.

Lemma 4.1. *For any $0 \le h \le h_{\max}$, given that $\delta(p^*, q) \le \tau\rho$, the distance between the exact nearest neighbor in layer $\mathcal{V}_h$ and the query point q satisfies:*

$$\delta(p_h^*, q) \le \psi_h \cdot R_h, \tag{4}$$

where $\psi_h = \sum_{k=0}^{h} \beta^{-k} + \frac{\tau}{\beta^h}$, and $R_h = \rho\beta^h$.

Proof. Let p_0^* denote the exact nearest neighbor at the bottom level (identical to the global nearest neighbor p^*). By Lemma 3.2 applied to p_0^*, there exists a covering vertex $u_h \in \mathcal{V}_h$ satisfying $\delta(u_h, p_0^*) \le \sum_{k=0}^{h} \rho\beta^k$. Combining this with the triangle inequality and our premise $\delta(p_0^*, q) \le \tau\rho$, we bound the distance from u_h to q:

$$\begin{aligned} \delta(u_h, q) \;&\le\; \delta(u_h, p_0^*) + \delta(p_0^*, q) \le \left(\sum_{k=0}^{h} \rho\beta^k\right) + \tau\rho \\ &= \rho\beta^h \left(\sum_{k=0}^{h} \beta^{k-h}\right) + \tau\rho = R_h \left(\sum_{j=0}^{h} \beta^{-j}\right) + R_h \left(\frac{\tau}{\beta^h}\right) \\ &= \left(\sum_{k=0}^{h} \beta^{-k} + \frac{\tau}{\beta^h}\right) R_h = \psi_h \cdot R_h. \end{aligned}$$

By definition, p_h^* is the strictly closest vertex to q in $\mathcal{V}_h$, dictating that $\delta(p_h^*, q) \le \delta(u_h, q) \le \psi_h \cdot R_h$. This establishes the desired upper bound, thereby completing the proof. □

Lemma 4.2. *For any $0 \le h < h_{\max}$, the relationship between the approximate nearest neighbor $\hat{p}_{h+1}$ at level h+1 and the exact nearest neighbor p_h^* at level h satisfies:*

$$\delta(\hat{p}_{h+1}, q) + \delta(p_h^*, q) \;\le\; \Delta_h \cdot R_h, \tag{5}$$

where $\Delta_h = \left(\frac{2\alpha}{\alpha-1}+\theta\right)\psi_h + \left(\frac{\alpha+1}{\alpha-1}+\theta\right)\beta$.

Proof. Let us denote the approximation factor as $\gamma = \frac{\alpha+1}{\alpha-1} + \theta$. Based on the approximation guarantee, we have $\delta(\hat{p}_{h+1}, q) \le \gamma\psi_{h+1}R_{h+1}$. By applying the recurrence relation $R_{h+1} = \beta R_h$ and observing the algebraic identity $\beta\psi_{h+1} = \psi_h + \beta$, the derivation smoothly unfolds:

$$\begin{aligned}\delta(\hat{p}_{h+1}, q) + \delta(p_h^*, q) &\le \gamma\psi_{h+1}R_{h+1} + \delta(p_h^*, q) \le \gamma\psi_{h+1}(\beta R_h) + \psi_h R_h \\ &= \gamma(\psi_h + \beta)R_h + \psi_h R_h = [(\gamma+1)\psi_h + \gamma\beta]\, R_h.\end{aligned}$$

Substituting γ back into the expression and recognizing the exact formulation of Δ_h, we conclude the proof. □

Building on Lemma 4.1 and Lemma 4.2, we present the main theorem regarding search accuracy guarantee. This theorem asserts that the combination of the hierarchical topology and the specific pruning logic of Artea Graph preserves sufficient connectivity to ensure correctness.

Theorem 4.3 (Accuracy Guarantee). *For any query q whose exact nearest neighbor p_0^* satisfies $\delta(p_0^*, q) \le \tau\rho$, greedy routing on $\mathcal{G}_h$ guarantees the following: for levels $h \ge 1$, it terminates at a high-quality vertex $\hat{p}_h$ satisfying $\delta(\hat{p}_h, q) \le \left(\frac{\alpha+1}{\alpha-1} + \theta\right)\psi_h R_h$; at the bottom level $h = 0$, it successfully converges to the exact NN p_0^*.*

Proof. We prove this proposition by induction on the level h, proceeding in a top-down manner, and the special scenario at the bottom level will be discussed in the induction step.

For the base case ($h = h_{\max}$), since the top level of the Artea Graph contains a single point $p^\top$, the routing process trivially returns $\hat{p}_{h_{\max}} = p^\top$, which coincides with $p^*_{h_{\max}}$ and serves as the entry point for the next level. By Lemma 4.1, $\delta(p^\top, q) \le \psi_{h_{\max}} R_{h_{\max}} \le \left(\frac{\alpha+1}{\alpha-1} + \theta\right)\psi_{h_{\max}} R_{h_{\max}}$, where the second inequality holds since $\frac{\alpha+1}{\alpha-1} + \theta > 1$ for any $\alpha > 1$ and $\theta \ge 0$. This establishes the base case.

For the induction step, we assume that greedy routing on level h+1 ($h \ge 0$) correctly returns a high-quality vertex $\hat{p}_{h+1}$ satisfying $\delta(\hat{p}_{h+1}, q) \le \left(\frac{\alpha+1}{\alpha-1}+\theta\right)\psi_{h+1}R_{h+1}$. Consequently, the routing on level h initializes at $\hat{p}_{h+1}$. Let p be the current vertex visited in $\mathcal{G}_h$. If the exact nearest neighbor (at this level) p_h^* is an out-neighbor of p, the algorithm naturally moves to p_h^*, ensuring correctness. Otherwise, p_h^* must have been removed from p's out-neighbor set $\mathcal{N}_h(p)$. We analyze this absence in two cases:

Case 1 (ARC Violation): p_h^* was excluded during the ARC-candidate construction (Algorithm 2, Line 1). By Lemma 4.2, we have:

$$\delta(p, p_h^*) \le \delta(p, q) + \delta(p_h^*, q) \le \delta(\hat{p}_{h+1}, q) + \delta(p_h^*, q) \le \Delta_h \cdot R_h,$$

where $\delta(p, q) \le \delta(\hat{p}_{h+1}, q)$ holds because routing on $\mathcal{G}_h$ starts from $\hat{p}_{h+1}$ (by the inductive hypothesis) and greedy routing monotonically reduce the distance to q. According to the construction rule, p_h^* satisfies the distance threshold and must have been included in the ARC-candidate set of p. Thus, we verify p_h^* is within the ARC-candidate range—such that its absence from the final neighbor set implies it must have been removed during the pruning phase, leading us to Case 2.

Case 2 (Pruning Rule Violation): p_h^* was pruned due to the pruning condition in Line 7. This implies the existence of out-neighbor p_{out} within the conflicting radius of p. We analyze this based on the level-specific pruning logic of Artea Graph.

(i) Upper Layers ($h \ge 1$). The pruning condition implies $\delta(p_{\text{out}}, p_h^*) \le \delta(p, p_h^*)/\alpha$. We proceed by contradiction. Suppose p is a local minimum but **not** satisfying $\delta(p, q) \le \left(\frac{\alpha+1}{\alpha-1}+\theta\right)\psi_h R_h$. By the triangle inequality and the pruning condition:

$$\begin{aligned}\delta(p_{\text{out}}, q) \;&\le\; \delta(p_{\text{out}}, p_h^*) + \delta(p_h^*, q) \;\le\; \frac{\delta(p, p_h^*)}{\alpha} + \delta(p_h^*, q) \\ &\le\; \frac{\delta(p, q)}{\alpha} + (1 + \frac{1}{\alpha}) \cdot \delta(p_h^*, q).\end{aligned} \tag{6}$$

By Lemma 4.1, we have, $\delta(p_h^*, q) \le \psi_h \cdot R_h \le \frac{\delta(p,q)}{\left(\frac{\alpha+1}{\alpha-1}+\theta\right)}$. Substituting the bound for $\delta(p_h^*, q)$, we derive:

$$\delta(p_{\text{out}}, q) < \delta(p, q)\left[\frac{1}{\alpha} + \left(1 + \frac{1}{\alpha}\right)\frac{1}{\left(\frac{\alpha+1}{\alpha-1}+\theta\right)}\right].$$

Provided that $\alpha > 1$ and $\theta > 0$, the term in the brackets is strictly less than 1. This yields $\delta(p_{\text{out}}, q) < \delta(p, q)$, contradicting the assumption that p is a local minimum. Thus, routing continues until a good enough $\hat{p}_h$ is found.

(ii) Bottom Layer ($h = 0$). At the bottom level, the pruning condition is relaxed using an offset $(\alpha + 1)\tau\rho$. A point p_0^* is pruned by p_{out} only if there is a neighbor p_{out} that occluded p_0^*, which implies $\delta(p_{\text{out}}, p_h^*) \le \frac{1}{\alpha}\left(\delta(p, p_h^*) - (\alpha + 1)\tau\rho\right)$.

We proceed by contradiction. Assume the current vertex p is a local minimum but is **not** the exact nearest neighbor (i.e., $p \ne p_0^*$). Since p_0^* is pruned in this case, the pruning condition at the bottom layer implies the existence of an out-neighbor p_{out} of p, such that $\delta(p_{\text{out}}, p_0^*) \le \frac{1}{\alpha}\left(\delta(p, p_0^*) - (\alpha + 1)\tau\rho\right)$. By combining this with the triangle inequality $\delta(p_{\text{out}}, q) \le \delta(p_{\text{out}}, p_0^*) + \delta(p_0^*, q)$, we derive:

$$\begin{aligned}\delta(p_{\text{out}}, q) \;&\le\; \delta(p_{\text{out}}, p_0^*) + \delta(p_0^*, q) \\ &\le\; \frac{1}{\alpha}\left(\delta(p, p_0^*) - (\alpha + 1)\tau\rho\right) + \delta(p_0^*, q) \\ &\le\; \frac{1}{\alpha}\left(\delta(p, q) + \delta(q, p^*) - (\alpha + 1)\tau\rho + \alpha\delta(p_0^*, q)\right) \\ &\le\; \frac{1}{\alpha}\left(\delta(p, q) - \alpha\tau\rho + \alpha\delta(p_0^*, q)\right) \le \delta(p, q)/\alpha\end{aligned} \tag{7}$$

Since $\alpha > 1$, this inequality implies $\delta(p_{\text{out}}, q) < \delta(p, q)$. This directly contradicts the assumption that p is a local minimum, as there exists a neighbor p_{out} strictly closer to q. Consequently, the routing cannot stop at any $p \ne p_0^*$ and must terminate at the exact NN.

Conclusion: The greedy routing algorithm is guaranteed to converge effectively. For upper layers ($h \ge 1$), the strict angular pruning enforces a monotonic descent until a $(1+\theta)$-approximation is reached; this approximation suffices as a high-quality entry point for the subsequent layer. In contrast, for the bottom layer ($h = 0$), the relaxed pruning logic effectively eliminates all false local minima, thereby guaranteeing that the routing terminates at the exact nearest neighbor p_0^*. □

4.2 Query Time Complexity

In this subsection, we demonstrate that the Artea Graph achieves superior query time complexity compared to all other proximity graphs with theoretical guarantees. Intuitively, the time complexity of ANN search on a PG is governed by two primary factors: the

routing length (i.e., the number of hops during traversal) and the out-degree of each vertex. Consequently, our theoretical analysis proceeds by proving that the routing path length is logarithmically bounded, and then deriving the total time complexity.

Proposition 4.4 (Routing Steps Bound). *The greedy routing on $\mathcal{G}_h$ terminates in $O(1)$ steps.*

Proof. Let $\pi = (v_0, v_1, \ldots, v_\ell)$ denote the sequence of vertices visited by the greedy routing at level h. The routing initializes at $v_0 = \hat{p}_{h+1}$, which is the entry point passed down from the upper level. As established in Theorem 4.3, the termination criterion differs by layer: for upper levels ($h \geq 1$), early termination is triggered once $\delta(v_\ell, q) \leq \gamma\psi_h R_h$ with $\gamma := \frac{\alpha+1}{\alpha-1} + \theta$; for the bottom level ($h = 0$), the routing converges to the exact nearest neighbor $v_\ell = p_0^*$. In both cases, our objective is to bound the path length ℓ.

We first establish a distance-reduction invariant. From the initial condition specific to our hierarchical routing and Lemma 4.1, the distance bound for v_0 expands as:

$$\delta(v_0, q) \leq \delta(\hat{p}_{h+1}, q) \leq \gamma \cdot \psi_{h+1} \cdot R_{h+1}. \tag{8}$$

Applying the inductive proof techniques from prior work [19, 24] to inequalities (6) and (7) under this initial condition (8), we deduce the following lemma:

Lemma 4.5. *For any level $0 \leq h < h_{\max}$ and any step $i < \ell$ during routing, the distance from the current vertex v_i to the query q satisfies:*

$$\delta(v_i, q) \leq \begin{cases} \dfrac{\gamma\,\psi_{h+1}\,R_{h+1}}{\alpha^i} + \dfrac{\alpha+1}{\alpha-1}\,\delta(p_h^*, q), & h \geq 1 \qquad (9) \\ \dfrac{\gamma\,\psi_1\,R_1}{\alpha^i}, & h = 0. \qquad (10) \end{cases}$$

With this recurrence in hand, we bound ℓ at each level.

Upper Levels ($h \geq 1$). The routing on $\mathcal{G}_h$ descends to $\mathcal{G}_{h-1}$ once $\delta(v_i, q) \leq \gamma\psi_h R_h$. Substituting Lemma 4.1, which gives $\frac{\alpha+1}{\alpha-1}\delta(p_h^*, q) \leq \frac{\alpha+1}{\alpha-1}\psi_h R_h$, into Eq. (9) yields

$$\delta(v_i, q) \;\leq\; \frac{\gamma\psi_{h+1}R_{h+1}}{\alpha^i} \;+\; \frac{\alpha+1}{\alpha-1}\psi_h R_h.$$

Decomposing the descent threshold as $\gamma\psi_h R_h = \frac{\alpha+1}{\alpha-1}\psi_h R_h + \theta\psi_h R_h$, it suffices to drive the transient term below the slack $\theta\psi_h R_h$:

$$\frac{\gamma\psi_{h+1}R_{h+1}}{\alpha^i} \leq \theta\psi_h R_h \Longleftrightarrow \alpha^i \geq \frac{\gamma\beta\psi_{h+1}}{\theta\psi_h} \overset{(\dagger)}{=} \frac{\gamma(\psi_h + \beta)}{\theta\psi_h}$$

where step (†) uses the algebraic identity $\beta\psi_{h+1} = \psi_h + \beta$. Since $\psi_h \geq 1$ (immediate from $\sum_{k=0}^{h} \beta^{-k} \geq 1$), the threshold is bounded by $\gamma(1+\beta)/\theta$, so $i \geq \log_\alpha(\gamma(1+\beta)/\theta) = O(1)$ steps suffice to trigger the early termination rule.

Bottom Level ($h = 0$). The greedy routing on the bottom level must terminate at p_0^*. The bottom-layer pruning rule enforces the *α-reducible property* (Theorem 4.3, Case 2(ii)): while $v_i \neq p_0^*$, $\delta(v_{i+1}, q) \leq \delta(v_i, q)/\alpha$. Iterating from $v_0 = \hat{p}_1$ with $\delta(v_0, q) \leq \gamma\psi_1 R_1$ yields Eq. (10). We split on the magnitude of $\delta(p_0^*, q)$.

Case A: $\delta(p_0^*, q) \geq \rho/2$. Setting $v_\ell = p_0^*$ in Eq. (10),

$$\ell \;\leq\; \log_\alpha \frac{\delta(v_0, q)}{\delta(p_0^*, q)} \;\leq\; \log_\alpha \frac{2\gamma\psi_1 R_1}{\rho}.$$

Case B: $\delta(p_0^*, q) < \rho/2$. By the r-net separation of $\mathcal{V}_0 = P$, every $v \in P \setminus \{p_0^*\}$ satisfies $\delta(v, p_0^*) \geq \rho$, so $\delta(v, q) > \rho/2$. Hence $\delta(v_{\ell-1}, q) > \rho/2$, and Eq. (10) gives

$$\ell \;\leq\; \log_\alpha \frac{2\delta(v_0, q)}{\rho} + 1 \;\leq\; \log_\alpha \frac{2\gamma\psi_1 R_1}{\rho} + 1.$$

As $\alpha, \beta, \gamma, \psi_1, \tau$ are constants, $\ell = O(1)$ in both cases.

Therefore, in both scenarios, the total path length is $\ell = O(1)$. □

By synthesizing Propositions 3.3 and 4.4, we determine the total time complexity of the routing procedure across all layers, thereby establishing the main theoretical result.

Theorem 4.6. *Let $\mathcal{G}$ be an Artea Graph constructed on a dataset P of size n. For any query point q with exact NN $p^* \in P$, if $\delta(q, p^*) \leq \tau\rho$ for a user-defined parameter $\tau > 0$, the hierarchical greedy routing on $\mathcal{G}$ guarantees retrieval of p^* in $O\left((\alpha\tau)^\lambda + \alpha^\lambda \log \Delta\right)$ time.*

5 ARTEALIB: ENHANCING PRACTICAL EFFICIENCY WITH ARTEA GRAPH

To accelerate Artea index construction while preserving search quality, we adopt a decoupled framework that separates the construction methodologies of the upper layers and the bottom layer. Since increment-based approaches require a massive candidate pool for neighbor selection, causing severe build-time bottlenecks [35, 46], we confine such incremental construction to the upper layers, whose cardinality is orders of magnitude smaller. There, we build an approximate dynamic R-net that efficiently handles insertions and exploits the evolving topology to determine layer-specific insertion criteria. In contrast, for the bottom layer encompassing all data points, we delegate the construction to a highly optimized, refinement-based paradigm that rapidly refines an initial graph into a search-efficient topology.

Increment-based Construction of Approximate Hierarchical r-Nets (Upper Layers). In the standard construction of hierarchical r-nets, to strictly enforce the covering and separation properties, a point p is promoted to layer h ($h > 0$) if and only if p satisfies $p \in \mathcal{V}_{h-1} \wedge \min_{v \in \mathcal{V}_h} \delta(p, v) > \rho\beta^h$. Since enforcing this exactness is computationally prohibitive, we propose a novel approach for building high-quality approximate hierarchical r-nets. Specifically, we relax the insertion condition, substituting the exact minimum with an approximate nearest neighbor obtained via graph routing:

$$p \in \mathcal{V}_{h-1} \wedge \delta(p, \mathsf{NNSearch}(C_{h+1}, p, \mathcal{G}_h)) > \rho\beta^h$$

Within this scheme, the upper layers are constructed by streaming pointwise insertion. To determine each point's insertion level h_p, we run a fast top-down BeamSearch with a moderate budget L_1, approximating the nearest-neighbor distance d_h^* at each layer. For the active insertion layers from h_p down to 1, we avoid redundant distance computations by warm-starting the search: reusing the candidate sets C_h cached during routing as seeds, BeamSearch resumes with an expanded budget L_2 for higher neighborhood quality. The expanded candidates are then distilled via ARC-Pruning (with layer-specific radius $\rho\beta^h$) to yield the neighborhood $\mathcal{N}_h(p)$.

Refinement-based Bottom Layer Construction. This phase establishes the bottom layer's massive edge set. Rather than the vertex-by-vertex insertion paradigm of the previous phase, we present an efficient graph construction framework that significantly reduces construction time. The core of our framework features an iterative refinement engine based on an efficient Bulk Synchronous Parallel

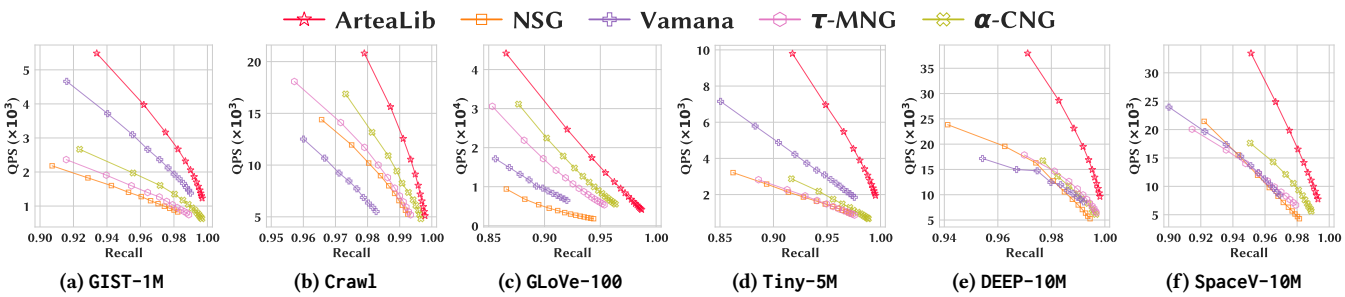


Figure 2: Recall@100 vs. throughput comparison between ArteaLib and flat indexes across different datasets.

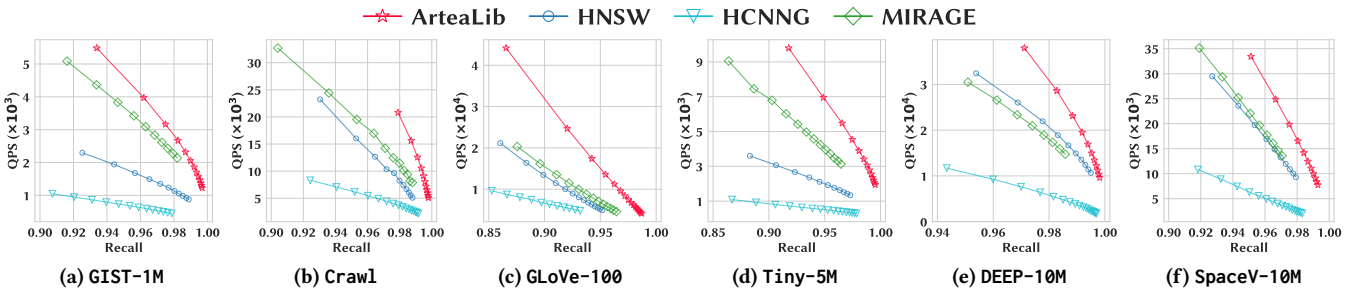


Figure 3: Recall@100 vs. throughput comparison between ArteaLib and hierarchical indexes across different datasets.

(BSP) model. In each iteration's computation phase, the engine assigns every vertex p to a thread, which applies a predefined *updater* to process p and its out-neighbors. Since some updaters also modify other vertices' out-neighborhoods, we use a log table to buffer such cross-vertex updates. In the synchronization phase, all vertices pull their pending updates from the log table, after which the table is flushed. By avoiding the locking overhead of read-write conflicts, this design accelerates construction while offering a clean abstraction for customizing refinement. We instantiate three updaters for bottom layer refinement:

Triangle Updater. Inspired by RNN-Descent, the intuition is that when a candidate v for p's neighborhood is pruned for being close to an existing neighbor u, v is a strong candidate to become an out-neighbor of u and can be recommended to u. The original RNN-Descent couples pruning (p, v) and recommending (u, v) under one coupled MRNG-based pruning condition. Tightening it to our novel pruning threshold $\mathcal{F}_{\text{AG}}$ makes pruning harder to trigger; such that fewer prunings yield fewer recommendations and a suboptimal graph. We therefore decouple the recommendation threshold from the pruning threshold, recovering the "missed" recommendations and yielding a well-connected graph.

Reverse Updater. For the processing vertex p, we log a reverse edge (u, p) for each out-neighbor u. These edges may meet the recommendation threshold, letting the Triangle Updater escape local optima and trigger further iterations; they also strengthen connectivity, improving recall for top-k queries.

ARC Updater. This updater enforces the Aspect Ratio Constraint of Section 2. As it mutates only processing vertex p's local state, it is thread-safe and free of contention overhead.

Orchestrating the three updaters yields the approximate Artea bottom layer: each of the n_2 outer rounds runs n_1 Triangle Updater self-loops followed by a single Reverse Updater pass; once all n_2 rounds complete, the ARC Updater finalizes the result. We fix $n_1 = 4$ and $n_2 = 15$ throughout all experiments.

6 EXPERIMENTS

6.1 Experimental Setup

Environment. Our experiments are conducted on a dual-socket server equipped with two Intel Xeon Gold 6326 CPUs @ 2.90GHz and 256GB DDR4 main memory. The system runs 64-bit Ubuntu 22.04.5 LTS, and all implementations are compiled using `g++14.3` with `-O3` optimization flag.

Datasets. We benchmark ArteaLib on eight real-world datasets (Table 2) spanning multiple modalities (image, text, and latent audio), with dimensionalities from 96 to 960 and scales from 1M to 10M. For datasets without predefined queries, we randomly hold out 10,000 base vectors as the query set, and compute exact ground truths via FAISS [8] brute-force ℓ_2 search.

Competing Methods. We compared Artea against seven state-of-the-art PG algorithms, grouped by overall topology. From the *hierarchical index* family, we included `HNSW` [31], `HCNNG` [45] and `MIRAGE` [46]. From the *flat index* family, we included `NSG` [11], `Vamana` [41], τ-`MG` [38], and α-`CG` [24]. Notably, we excluded `RNN-Descent` [35] from our baselines because `MIRAGE` incorporates it as its bottom layer and has been demonstrated to consistently outperform it [46]. All baseline implementations used publicly available source code. For each baseline and dataset, we adopted the parameter settings recommended in the original papers [24, 46] or previous benchmark studies [47] whenever available; the search-time parameter (i.e. beam width) was always swept to trace the

Table 2: Summary of Datasets.

Dataset	Dim	#Base	#Query	Source	Type
SIFT-1M	128	1,000,000	10,000	[20]	Image
GIST-1M	960	1,000,000	1,000	[20]	Image
Crawl	300	1,998,995	10,000	[32]	Text
GloVe-100	100	1,286,614	10,000	[39]	Text
YahooMusic	300	1,822,179	1,000	[1]	Latent
Tiny-5M	384	5,000,000	1,000	[1]	Image
DEEP-10M	96	10,000,000	29,316	[48]	Image
SpaceV-10M	100	10,000,000	10,000	[6, 40]	Text

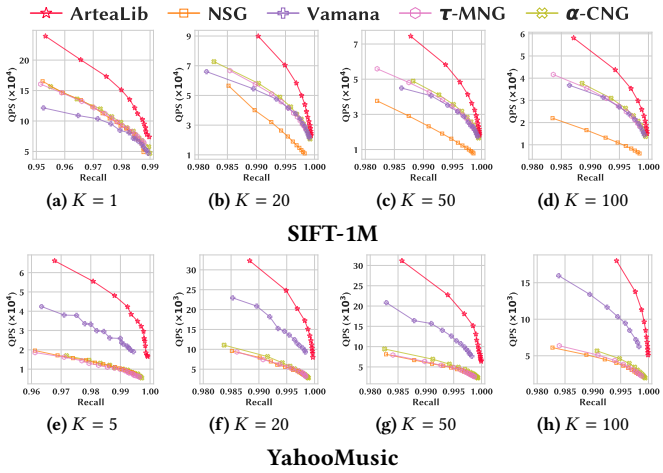


Figure 4: Recall@K vs. throughput comparison between ArteaLib and flat indexes across varying K.

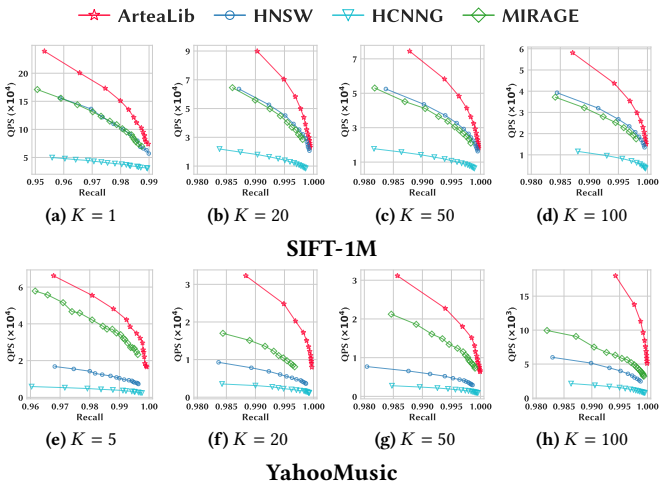


Figure 5: Recall@K vs. throughput comparison between ArteaLib and hierarchical indexes across varying K.

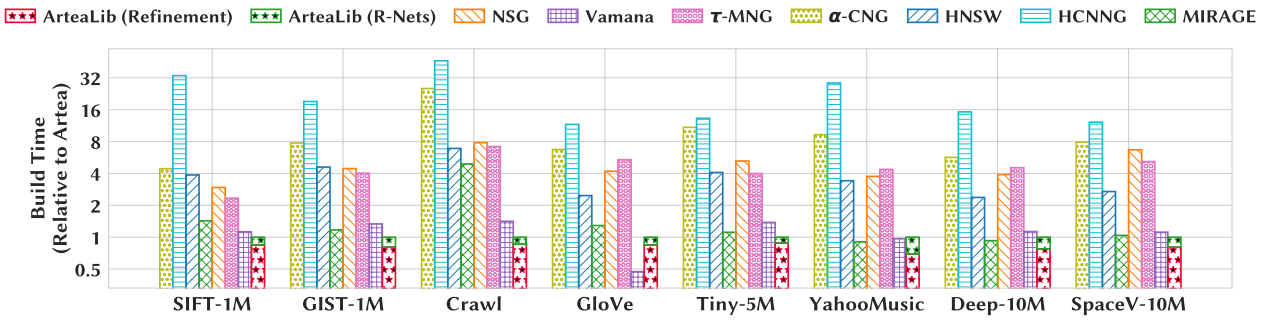


Figure 6: Index construction time comparison between ArteaLib and the baselines across all evaluated datasets.

full recall–QPS Pareto frontier. For index construction, we repeated the experiments three times, while for search performance, each configuration was evaluated 20 times.

6.2 Search Performance

In this section, we evaluated the end-to-end search performance of our algorithm against other baselines. We employed Recall@K vs. QPS (Queries Per Second) as the primary evaluation metric, with all algorithms executed in parallel using the maximum number of system threads. To plot the performance curves, we varied the runtime search parameter—specifically, the candidate queue budget—to collect diverse Recall-QPS data points. A higher QPS at a given recall level typically indicates that fewer distance computations are required during graph traversal. Consistent with prior work [24], our primary evaluation focuses on $K = 100$. For a more comprehensive comparison across diverse top-K retrieval scenarios, we also vary K and report the recall at $K \in \{1, 5, 20, 50, 100\}$.

To avoid overcrowded curves and to better illustrate the impact of the graph structure, we split the comparison against baselines into two groups: flat indexes and hierarchical indexes, as shown in Figures 2–5. To provide a more intuitive comparison, we set a target recall for each dataset according to its difficulty and reported the QPS that each baseline achieved at this target recall. The results demonstrate that ArteaLib consistently outperforms all baseline solutions across all datasets and experimental configurations. To concisely quantify this advantage, Table 3 summarizes the average and maximum QPS speedups achieved by ArteaLib over each competing method under the standard K=100 setting.

Table 3: QPS speedup of ArteaLib over baseline methods.

Speedup	Hierarchical Family			Flat Family			
	HNSW	HCNNG	MIRAGE	NSG	Vamana	τ-MG	α-CG
Average	2.22×	6.19×	1.56×	3.31×	2.07×	2.38×	1.97×
Maximum	3.50×	13.60×	2.20×	8.46×	3.85×	4.70×	3.86×

6.3 Index Construction Performance

Construction Time. To clearly illustrate the relative efficiency, we normalize the construction time of Artea to 1 across all datasets, and report the execution time of other baselines as a relative factor (speedup or slowdown) compared to Artea. As depicted in Figure 6, the two hierarchical index structures (i.e., HNSW and HCNNG) typically demand substantially longer construction periods. Meanwhile, several proximity graphs backed by theoretically sound prototypes also suffer from heavy indexing overhead due to their complex pruning or selection phases. In contrast, Artea demonstrates remarkably fast construction speeds, achieving the optimal indexing efficiency in most cases. This superior performance primarily stems from the highly optimized infrastructure implemented in ArteaLib and our efficient construction strategy selection. Additionally, as

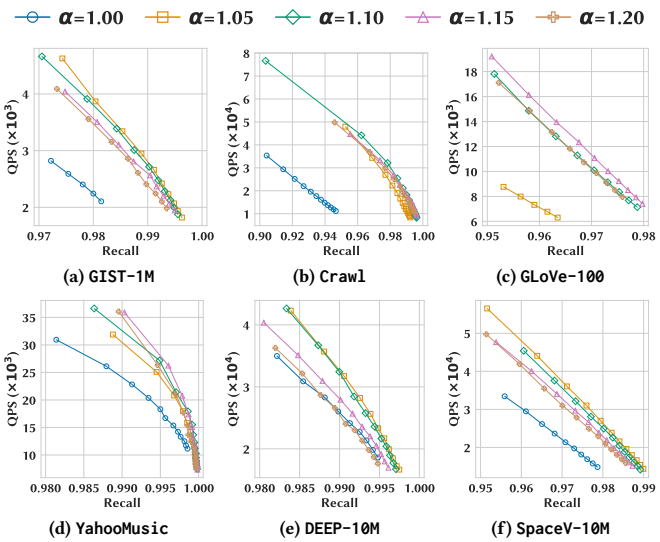


Figure 7: The impact of varying α on search performance.

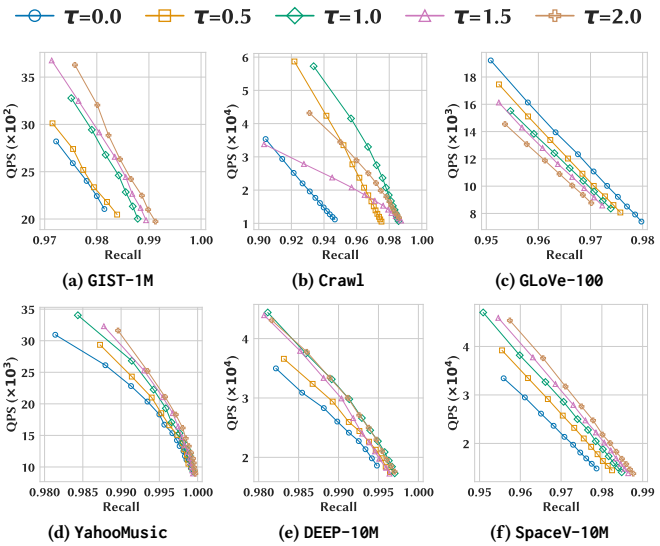


Figure 8: The impact of varying τ on search performance.

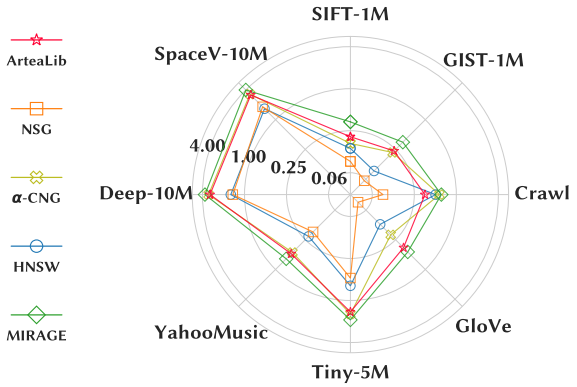


Figure 9: Index size (GB) comparison of ArteaLib against representative baselines.

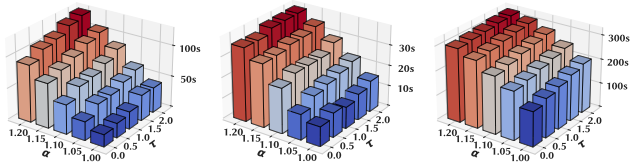


Figure 10: The impact of varying α and τ on build time.

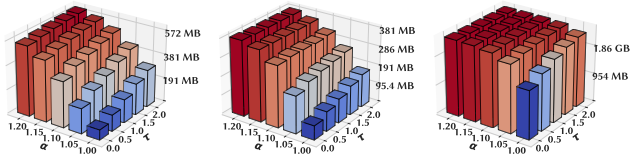


Figure 11: The impact of varying α and τ on index size.

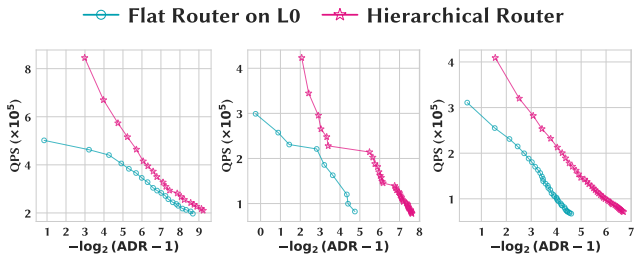
distinguished by the colors in Artea's bars, constructing the upper layers (i.e., approximate r-nets) consumes only a minor portion of the overall construction time.

Index Size. Regarding the memory overhead, we observe that Artea exhibits a larger index size compared to some flat baseline graphs. This is mainly attributed to its multi-layer hierarchical architecture; as illustrated in Figure 9, hierarchical graphs naturally require more edges to maintain connectivity across different layers. Even so, Artea remains more space-efficient than existing hierarchical baselines like MIRAGE. Nevertheless, given that proximity graphs typically occupy only a small fraction of the space required by raw vectors, we argue that this slight increase in index size is a highly worthwhile trade-off to unlock significantly superior QPS versus Recall Pareto performance.

6.4 Effect of Parameters

This subsection explores the effects of α and τ in our pruning scheme (Section 3.2). Unlike the evaluation frameworks of prior works, our parameter study investigates the impact of α and τ not only on search performance, but also on index construction time and index size. This provides a more comprehensive understanding of their practical implications. We varied α from 1.00 to 1.20 and τ from 0.0 to 2.0 for each dataset.

Impact on Search Performance. To evaluate the search performance across different parameter combinations, we tuned the runtime candidate queue size to reach a predefined target recall@20, and subsequently recorded the corresponding Queries Per Second (QPS) for comparative analysis. For configurations that failed to achieve the target recall even under the maximum candidate queue budget, we explicitly mark their corresponding bars with a cross. Consistent with observations in prior studies, indefinitely increasing α and τ does not yield continuous gains. Intuitively, for a fixed degree bound, larger values of α and τ relax the pruning conditions, preserving not only more neighbors but also a higher proportion of short-range edges. Consequently, scaling these parameters introduces a nuanced trade-off between search efficiency and query accuracy: while the computational cost per routing step increases, the improved graph quality enables us to achieve higher recall under the same candidate queue budget. Empirically, we observe that optimal performance is almost universally achieved at moderate parameter settings (e.g., $\alpha = 1.10$ and $\tau = 1.5$).



(a) SIFT-1M (b) DEEP-10M (c) SpaceV-10M

Figure 12: Ablation Study: ADR@1 vs. QPS between ARTEA and ARTEA Layer 0.

Impact on Construction Time and Index Size. Regarding index size and construction time, in contrast to the trends observed in search performance, increasing α and τ generally leads to an inflation in both metrics. This occurs because the pruning condition becomes progressively more conservative as these parameters grow, thereby retaining more edges in the graph. Consequently, the total number of edges evaluated during our refinement algorithm increases significantly. Therefore, the overhead in index size and construction time represents the inherent trade-off for utilizing larger parameter values to achieve superior search performance.

6.5 Ablation Study

To further understand the effect of ARTEA's hierarchical structure, we compare the full ARTEA Graph against its flat counterpart, obtained by retaining only the bottom layer. For the flat variants, queries begin from randomly chosen entry points, since no upper-layer descent is available to seed the search; the full ARTEA Graph, in contrast, inherits its entry point from the routing on the layers above. We report throughput (QPS) against retrieval quality for top-1 search, tracing each method's trade-off curve by sweeping the beam-search candidate queue size L. Quality is measured by the Average Distance Ratio (ADR ≥ 1); the x-axis plots $-\log_{10}(\text{ADR}-1)$ so that larger values correspond to higher accuracy.

As shown in Figure 12, the hierarchical design provides a clear performance benefit. Even at matched maximum out-degree, ARTEA Graph achieves higher QPS than ARTEA-Flat throughout the accuracy range, confirming that the hierarchical structure of ARTEA meaningfully accelerates the routing.

7 RELATED WORK

While various techniques—such as hashing [12, 18, 26, 29, 43, 44], inverted indices [13, 33, 42], and tree structures [4, 5, 21, 22]—address the ANN problem, proximity graphs (PGs) [9–11, 16, 24, 28, 30, 31, 35, 38, 41, 45, 46] dominate due to consistently superior performance. PGs locate approximate nearest neighbors by navigating a directed graph whose edges are constructed based on specific *predefined criteria*. Recent investigations into theoretically sound PGs [11, 19, 24, 28, 38] broadly classify them into two categories: *pruning-based* and *structural-driven* approaches.

Pruning-Based Approach. Methods in this category construct PGs through edge pruning strategies [11, 19, 24, 38], where formal variations in pruning criteria yield fundamentally different theoretical guarantees. The monotonic relative neighborhood graph (MRNG) [11] relies on the standard triangle inequality; however, it suffers from an $O(n)$ worst-case complexity and, more critically, fails to provide any search guarantees when $q \notin P$. To address this limitation, the τ-monotonic graph (τ-MG) [38] introduces a shifted term into the triangle inequality. Under the practical setting where the distance between a query q and its exact nearest neighbor v^* is strictly bounded by some constant, τ-MG guarantees exact retrieval; however, its worst-case time complexity is still bounded by $O(n)$. Subsequently, Indyk and Xu [19] demonstrated that the theoretical prototype of Vamana employs a scaled term in its pruning inequality. This adaptation allows Vamana to guarantee a $\left(\frac{\alpha+1}{\alpha-1}+\epsilon\right)$-approximate nearest neighbor (ANN) even for out-of-sample queries, establishing a poly-logarithmic worst-case search complexity of $O\left(\alpha^\lambda \log\Delta \log\frac{\Delta}{(\alpha-1)\epsilon}\right)$. Most recently, the α-convergent graph (α-CG) [24] synergized the scaled term from Vamana with the shifted term from τ-MG. Operating under the same practical setting, α-CG guarantees exact nearest neighbor retrieval within a worst-case complexity of $O\big((\alpha\cdot\tau)^\lambda \log\Delta \log_\alpha \Delta\big)$. While these methods meticulously design pruning inequalities to derive stronger accuracy guarantees and tighter query complexities, they are strictly confined to flat structures. Breaking this barrier, ARTEA extends the rigorous pruning paradigm to complex hierarchical structures, achieving a significant improvement in query complexity.

Structural-Driven Approach. The second category constructs proximity graphs by directly exploiting the spatial properties of r-nets [22], which does not rely on pruning-based paradigm, represented by G_{net} [28]. It directly leverages the r-net data structure, constructing multiple r-nets and establishing edges between each vertex v and points within specific ranges across these nets. By directly exploiting the structural properties of r-nets, G_{net} bypasses the edge pruning step, introducing a different methodology for fast proximity graph construction. Additionally, HENN [7] considers ϵ-nets for analysis; however, it lacks rigorous accuracy guarantees and fails to yield stronger time complexity bounds.

About r-Nets. G_{net} is the closest theoretical work to ours in its use of r-nets. While both ARTEA and G_{net} draw inspiration from r-nets, their designs differ fundamentally in graph construction methods and routing strategies. ARTEA enforces edge pruning within each layer and performs *hierarchical* greedy search, whereas G_{net} uses these nets mainly as a structural device for graph construction and still performs *flat* greedy routing. Consequently, G_{net}'s query complexity retains a polylogarithmic dependence on the global aspect ratio Δ. Note that earlier tree-based methods [5, 22] also utilized r-nets, but their routing mechanisms differ significantly from graph routing, placing them outside the scope of our discussion.

8 CONCLUSION

This paper presents ARTEA, a principled hierarchical proximity graph with rigorous accuracy guarantees and low time complexities. By systematically co-designing a deterministic r-net vertex hierarchy with our novel ARC-PRUNING strategy, ARTEA elegantly bounds the per-layer out-degree to a constant. This structural innovation makes ARTEA the first proximity graph to achieve a strictly logarithmic search complexity with respect to the dataset aspect ratio Δ, under practical assumptions. Furthermore, we translate the theoretical insights into a practical approach ARTEALIB, achieving state-of-the-art empirical performance.

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A PROOF OF PROPOSITION 3.3

Proof. In this section, we verify the space complexity (out-degree bound) stated in Proposition 3.3. Consider an arbitrary point p at layer h. We aim to bound the cardinality of its neighbor set $\mathcal{N}_h(p)$. The search range is bounded by $R_{\text{search}} = \Delta_h R_h$. We partition the search space around p into a sequence of disjoint concentric annuli $\mathcal{A}_i$, where the i-th annulus has an inner radius $\xi_i = 2^i R_h$ and covers the range $(\xi_i, 2\xi_i]$. The minimum pairwise separation σ_i within $\mathcal{A}_i$ is governed by the R_h-net property ($\sigma_i \geq R_h$) and the pruning condition defined in Eq. (3). We analyze the cardinality by distinguishing between upper layers and the bottom layer.

Scenario 1: Upper Layers ($h \geq 1$). For $h \geq 1$, the pruning condition requires $\delta(u, v) > \delta(p, u)/\alpha$. Thus, the separation in annulus $\mathcal{A}_i$ is bounded by:

$$\sigma_i \geq \max\left\{R_h, \frac{\xi_i}{\alpha}\right\}.$$

(i) Near Field ($\xi_i \leq \alpha R_h$): Consider the range where $\xi_i \leq \alpha R_h$. Here, the separation is dominated by R_h. The points of all these annulus reside in a volume of radius $O(\alpha R_h)$ with separation R_h. By the Lemma 2.3, the cardinality is:

$$|\mathcal{N}_h^{\text{near}}| \leq O\left(\left(\frac{\alpha R_h}{R_h}\right)^{\lambda}\right) = O(\alpha^{\lambda}).$$

(ii) Far Field ($\xi_i > \alpha R_h$): Consider the range where $\xi_i > \alpha R_h$. Here, $\sigma_i \geq \xi_i/\alpha$. The aspect ratio of the annulus region is constant ($O(\alpha)$), meaning each annulus contributes at most $O(\alpha^{\lambda})$ neighbors. Summing over the logarithmic range of annuli (bounded by $\log \Delta_h$):

$$|\mathcal{N}_h^{\text{far}}| \leq \sum O(\alpha^{\lambda}) = O(\alpha^{\lambda} \log \Delta_h).$$

Result ($h \geq 1$): Summing both fields yields $|\mathcal{N}_h(p)| = O(\alpha^{\lambda} \log \Delta_h)$.

Scenario 2: Bottom Layer ($h = 0$). For $h = 0$, the pruning condition is relaxed with an offset $C_0 = (\alpha + 1)\tau\rho$. Assuming the density parameter $\rho = \Theta(R_0)$, the separation bound becomes:

$$\sigma_i \geq \max\left\{R_0, \frac{1}{\alpha}(\xi_i - C_0)\right\}.$$

(i) Near Field ($\frac{1}{\alpha}(\xi_i - C_0) \leq R_0$): The Near Field extends as long as the pruning term is weaker than the intrinsic resolution, i.e., $\frac{1}{\alpha}(\xi_i - C_0) \leq R_0$. This implies:

$$\xi_i \leq \alpha R_0 + C_0 = \alpha R_0 + (\alpha + 1)\tau\rho = O(\alpha(1 + \tau)R_0).$$

In this expanded Near Field, points are separated by R_0 within a radius of $O(\alpha(1 + \tau)R_0)$. Applying the Packing Lemma:

$$|\mathcal{N}_0^{\text{near}}| \leq O\left(\left(\frac{\alpha(1 + \tau)R_0}{R_0}\right)^{\lambda}\right) = O\left((\alpha(1 + \tau))^{\lambda}\right).$$

(ii) Far Field ($\frac{1}{\alpha}(\xi_i - C_0) > R_0$): When ξ_i exceeds the Near Field boundary, the offset C_0 becomes negligible relative to ξ_i, and the separation scales as $\sigma_i \approx \xi_i/\alpha$. Similar to the upper layers, the count per annulus is $O(\alpha^{\lambda})$. Summing over the remaining layers yields $O(\alpha^{\lambda} \log \Delta_0)$.

Result ($h = 0$): The total cardinality is dominated by the expanded Near Field density and the logarithmic span. Summing both fields, and using $(1 + \tau)^{\lambda} = O(1 + \tau^{\lambda})$ together with $\log \Delta_0 = \Theta(1)$ (so the residual α^{λ} term is absorbed into $\alpha^{\lambda} \log \Delta_0$), we obtain:

$$|\mathcal{N}_0(p)| = O\left((\alpha\tau)^{\lambda} + \alpha^{\lambda} \log \Delta_0\right).$$

Conclusion. The out-degree is strictly bounded in both upper and bottom levels. For upper layers, it scales with $O(\alpha^{\lambda} \log \Delta_h)$, while for bottom layer, the relaxed pruning introduces an additive $(\alpha\tau)^{\lambda}$ term, resulting in $O((\alpha\tau)^{\lambda} + \alpha^{\lambda} \log \Delta_0)$. □